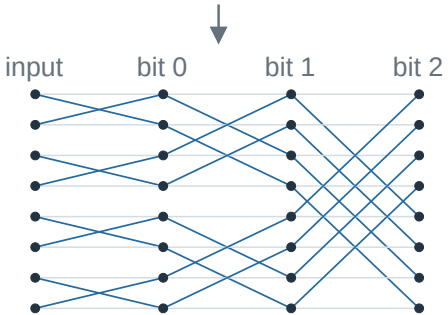
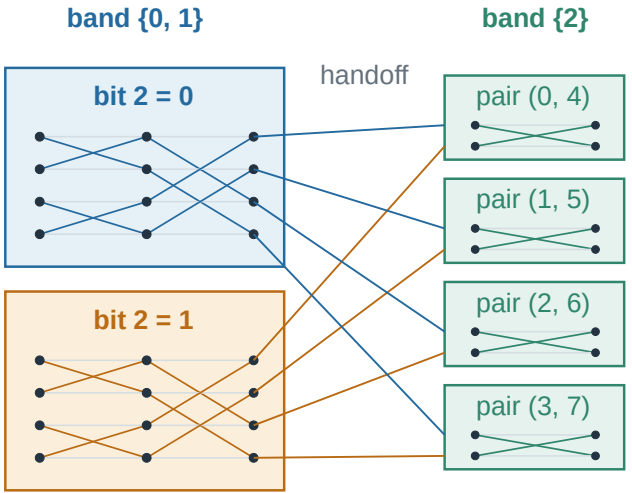
(a) Labeled dependencies

- **FFT** a +/- twiddle * b
- **NTT** a +/- twiddle * b (mod q)
- **FWHT** a + b, a - b
- **zeta** a, a + b
- **structured** M [a, b]

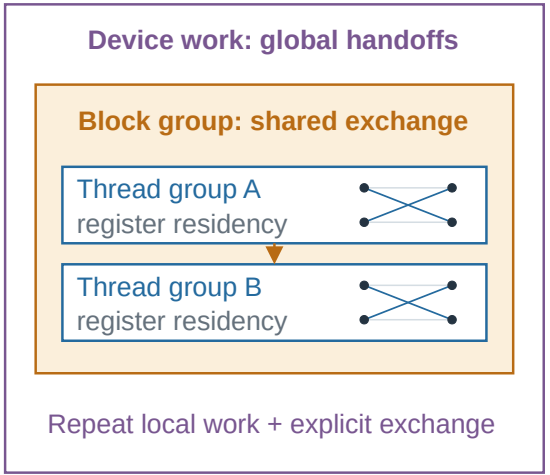


N = 8; common pairing envelope

(b) Bounded bands and fan-in



(c) Recursive hardware placement



tile 0 | tile 1 | tile 2

Independent tiles reuse resources